Uppala Surya Teja

es24btech11038@iith.ac.in | suryatejauppala9@gmail.com | +91 6303678250 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.39
SR Junior College	Class XII (Telangana State Board of Intermediate Education)	2024	98.60%
Hymavathi High School	Class X (Telangana State Board)	2022	9.80%

SKILLS

Programming Languages: C, C++, Python, SQL, JavaScript

Libraries & Frameworks: NumPy, Pandas, FastAPI, TensorFlow, Keras, Scikit-learn, XGBoost, Flask, Librosa, CVXPY

Tools & Technologies: Git, OSQP Solver, LaTeX, Streamlit, VS Code, Google Colab

Competitive Programming & DSA: Codeforces (Max Rating: **1556**), **500+** problems on LeetCode

PROJECTS

Quantitative Portfolio Risk & Analytics Platform | *Python, NumPy*

- Built a fully vectorized Monte Carlo VaR/CVaR engine generating 10,000+ simulation paths as a single NumPy matrix operation, eliminating loop-based bottlenecks for near-instant computation at scale
- Validated risk models via rolling 252-day out-of-sample backtesting over ~1,000 trading days, achieving 95.12% empirical accuracy and passing Kupiec's POF statistical test (p = 0.87)
- Designed a stress-testing framework and interactive visualizations for rolling market exposure and pairwise correlations, supporting exploratory data analysis
- Integrated the Gemini API to synthesize 19 quantitative metrics into automated, plain-English analytical commentary

Pricket - IPL Match Intelligence Platform | *Python, FastAPI, XGBoost, HTML/CSS/JS*

- Designed and built a full-stack, production-style system end-to-end, 11 REST API endpoints (FastAPI) spanning live simulation streaming, prediction, and squad-optimization across a 7,400+ line codebase
- Built an ETL pipeline processing 288K+ rows of 17-year ball-by-ball data into ML-ready splits, training an XGBoost win-prediction model at 65% accuracy on 1,100+ matches using 30 engineered features
- Engineered a NumPy-optimized Monte Carlo engine executing 500 full-match simulations per request, utilizing FastAPI Server-Sent Events (SSE) to stream real-time score distributions and prediction intervals to a zero-reload dashboard with sub-second latency.
- Eliminated look-ahead bias through strict train/serve data separation, ensuring model correctness under real-time inference load
- Implemented Ideal XI and Dream11 Fantasy XI recommendation engines using matchup statistics and position-aware selection, exposed through modular REST APIs

Multimodal Speech Emotion Recognition | *Python, TensorFlow*

- Engineered a high-throughput audio pipeline extracting 128-bin Mel spectrograms with augmentation (time-stretching, pitchshifting, noise injection) on the RAVDESS dataset (1,440 clips, 24 speakers)
- Built a multimodal architecture fusing a custom CNN audio branch with a Whisper-transcribed, Bidirectional-GRU text branch, evaluating Early and Late Fusion strategies
- Achieved 84.84% classification accuracy across 8 emotional states with 5.7ms average inference latency per clip on a Tesla T4 GPU

Producer-Consumer Synchronization System | *C, POSIX Threads*

- Designed and implemented a bounded-buffer producer-consumer system in C using POSIX threads, mutex locks, and condition variables to synchronize concurrent access to a shared circular queue with zero data races or busy-waiting
- Extended the design into a multi-process architecture using fork(), System V shared memory, and semaphore sets, coordinating two producers and one consumer safely across independent address spaces
- Applied core OS principles like mutual exclusion, deadlock avoidance, and inter-process communication to build a correct, race-free concurrent system from first principles

POSITIONS OF RESPONSIBILITY

Office of Career Services | Career Cell Coordinator

- Educated students on diverse career opportunities and coordinated campus placement logistics.

Sunshine (Counselling Cell) | Mentor

- Mentored juniors on stress management and organized campus campaigns to destigmatize mental health.

Finance and Consulting Club | Corporate Growth & Partnerships Coordinator

- Managed corporate outreach, secured sponsorships, and coordinated industry-led finance workshops.

Inter IIT Sports Meet | Public Relations Coordinator

- Directed **PR strategies**, managed official communications, and led **digital outreach** to maximize audience engagement.

RELEVANT COURSES

Computer Science: Algorithms, Data Structures & Applications, Theory of Computation, Operating Systems (OS), Database Management Systems (DBMS)

Mathematics: Probability & Random Variables, Statistics, Linear Algebra, Numerical Methods, Number Theory, Discrete Mathematics

EXTRACURRICULAR

- Achieved a **Rapid rating of 1296** on Chess.com.
- Secured the **Silver Medal in Chess** at Milan '24 (Annual Inter-Hostel Championship, **IIT Hyderabad**).